

If a linear system is consistent, then the solution set contains either

- a unique solution, when there are no free variables, or
- infinitely many solutions, when there is at least one free variable. $\square$

The following procedure outlines how to find and describe all solutions of a linear system.

Using Row Reduction to Solve a Linear System

- Write the augmented matrix of the system.
- Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent. If there is no solution, stop; otherwise, go to the next step.
- Continue row reduction to obtain the reduced echelon form.
- Write the system of equations corresponding to the matrix obtained in step 3.
- Rewrite each nonzero equation from step 4 so that its one basic variable is expressed in terms of any free variables appearing in the equation.

Practice Problems

- Find the general solution of the linear system whose augmented matrix is

$$\left[\begin{array}{ccc|c} \boxed{1} & -3 & -5 & 0 \\ 0 & \boxed{1} & -1 & -1 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -8 & -3 \\ 0 & 1 & -1 & -1 \end{array} \right]$$

Soln: 1. Row 1 = Row 1 + 3x Row 2

The corresponding equations are:

$$\begin{aligned} x_1 - 8x_3 &= -3 & (1) \\ x_2 - x_3 &= -1 & (2) \end{aligned}$$

The basic variables are x_1, x_2 . The free variables are x_3 . $\therefore$ The general soln. is:

$$\begin{cases} x_1 = 8x_3 - 3 \\ x_2 = x_3 - 1 \\ x_3 \text{ is free} \end{cases}$$